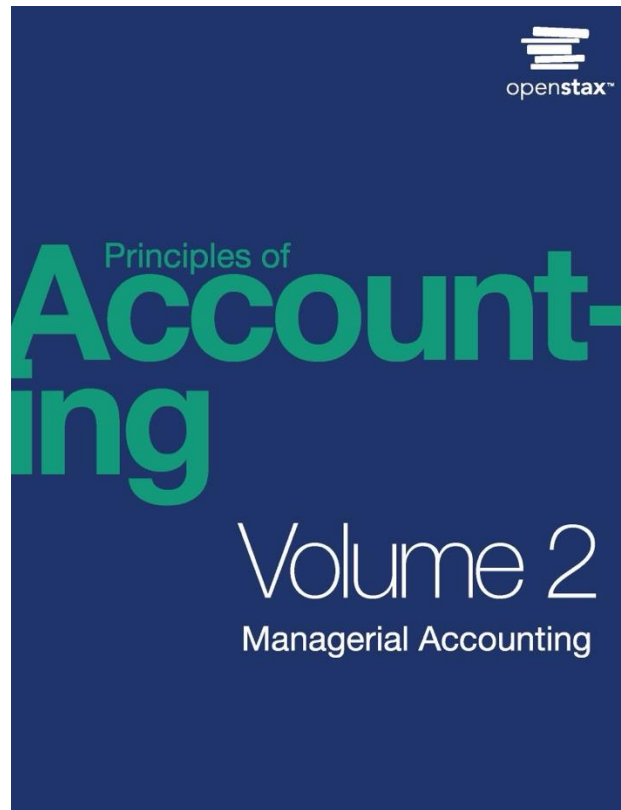


Principles of Accounting, Volume 2: Managerial Accounting

Chapter 6 ACTIVITY-BASED, VARIABLE, AND ABSORPTION COSTING

PowerPoint Image Slideshow



Chapter Outline

- 6.1 Calculate Predetermined Overhead and Total Cost under the Traditional Allocation Method
- 6.2 Describe and Identify Cost Drivers
- 6.3 Calculate Activity-Based Product Costs
- 6.4 Compare and Contrast Traditional and Activity-Based Costing Systems
- 6.5 Compare and Contrast Variable and Absorption Costing

6.1 Calculate Predetermined Overhead and Total Cost under the Traditional Allocation Method

- Component categories under traditional allocation
 - $\text{Product or job cost} = \text{direct materials} + \text{direct labor} + \text{manufacturing overhead}$
- Estimated total manufacturing overhead costs
 - Overhead application rate based on budgeted overhead costs and a budgeted level of activity

Figure 6.3

The materials and labor costs per unit for three different recorders produced by Musicality Manufacturing:

	Solo	Band	Orchestra
Direct Materials per Unit	\$ 3.50	\$6.00	\$11.70
Direct Labor Cost per Unit	10.00	2.75	4.30

Figure 6.4

Musicality's overhead per unit using traditional allocation:

	Solo	Band	Orchestra	Total
Direct Labor Hours per Product	350,000	400,000	500,000	1,250,000
Overhead Rate per Direct Labor Hour				
Overhead Assign per Product				
Number Units	140,000	100,000	250,000	490,000
Overhead per Unit	\$			

Estimated Annual Overhead Costs

\$2,500,000

Estimated Annual Activity

1,250,000 direct labor hours

Figure 6.5

Musicality's per unit product costs using traditional allocation:

	Solo	Band	Orchestra
Direct Materials per Unit			
Direct Labor per Unit			
Overhead per Unit	_____	_____	_____
Cost per Unit	=====	=====	=====

Musicality's gross profit per unit using traditional allocation:

	Solo	Band	Orchestra
Sales Price	\$20.00	\$25.00	\$30.00
Cost per Unit (traditional)			
Gross Profit per Unit			

Figure 6.6

Figure 6.7

Musicality's gross profit by product line using traditional allocation:

Traditional Costing	Solo	Band	Orchestra	Total
Number of Units Sold	150,000	110,000	200,000	460,000
Sales	\$3,000,000	\$2,750,000	\$6,000,000	\$11,750,000
Less Cost of Goods Sold				
Direct Material	525,000	660,000	2,340,000	3,525,000
Direct Labor	1,500,000	302,500	860,000	2,662,500
Manufacturing Overhead	750,000	880,000	800,000	2,430,000
Cost of Goods Sold	<u>2,775,000</u>	<u>1,842,500</u>	<u>4,000,000</u>	<u>8,617,500</u>
Gross Profit	<u>\$ 225,000</u>	<u>\$ 907,500</u>	<u>\$2,000,000</u>	<u>\$ 3,132,500</u>

Chapter Outline

6.1 Calculate Predetermined Overhead and Total Cost under the Traditional Allocation Method

6.2 Describe and Identify Cost Drivers

6.3 Calculate Activity-Based Product Costs

6.4 Compare and Contrast Traditional and Activity-Based Costing Systems

6.5 Compare and Contrast Variable and Absorption Costing

6.2 Describe and Identify Cost Drivers

- **Cost driver:** the reason a cost occurs
 - In a labor-intensive environment, direct labor hours or \$ are a driver of cost.
 - In a machine intensive environment, machine hours are a cost driver.
- Identify cost drivers:

Common Manufacturing Expenses and Potential Cost Drivers

Common Expenses	Potential Cost Drivers
• Customer Service • Cleaning Equipment Costs • Marketing Expenses • Office Supplies • Green Floral Tape (indirect material) • Website Maintenance Expense	• Number of product returns from customers • Number of square feet • Number of customer contacts • Number of employees • Number of customer orders • Number of customer online orders

Table 6.1

Chapter Outline

6.1 Calculate Predetermined Overhead and Total Cost under the Traditional Allocation Method

6.2 Describe and Identify Cost Drivers

6.3 Calculate Activity-Based Product Costs

6.4 Compare and Contrast Traditional and Activity-Based Costing Systems

6.5 Compare and Contrast Variable and Absorption Costing

6.3 Calculate Activity-Based Product Costs

- Historical perspective on determination of manufacturing overhead allocation
 - Much of production historically was labor based, making labor hours or labor dollars a good match for overhead allocation.
 - Increased use of machines makes machine hours an appropriate cost driver.
 - Both labor hours and machine hours are easily measured.
- Establishing an activity-based costing system
 - Technology has allowed for more detailed measurement of the many costs that make up overhead in many industries.

The Calculation of Product Costs Using the Activity-Based Costing Allocation Method

Step 1: Identify cost pools

Step 2: Assign overhead costs to cost pools

Step 3: Identify the cost drivers for each activity, and estimate annual activity levels

Step 4: Compute the predetermined overhead rate for each cost driver

Step 5: Allocate overhead costs to products to get the total overhead and overhead per unit

The Calculation of Product Costs Using the Activity-Based Costing Allocation Method



Step 1: Identify cost pools

- Musicality is considering using activity-based costing and has determined its cost pools:
 - 1) Setting up machines
 - 2) Purchasing materials
 - 3) Inspecting products
 - 4) Assembling products
 - 5) Technological production

Calculation of Product Costs Using ABC Allocation Method

Step 1: Identify cost pools

Step 2: Assign overhead costs to cost pools

Activity	Estimated Overhead Costs
Setting Up Machines	\$ 200,000
Purchasing Material	500,000
Inspecting Products	300,000
Assembling Products	600,000
Technological Production	900,000
Total	<u>\$2,500,000</u>

Calculation of Product Costs Using ABC Allocation Method

Step 1: Identify cost pools

Step 2: Assign overhead costs to cost pools

Step 3: Identify the cost drivers for each activity and estimate annual activity levels.

Cost Driver	Expected Cost Driver Activities			Total
	Solo	Band	Orchestra	
Machine Setups	2,000	1,500	1,500	5,000
Number of Purchase Requisitions	5,000	4,000	1,000	10,000
Inspection Hours	10,000	9,000	1,000	20,000
Number of Parts Requiring Labor	15,000	3,000	12,000	30,000
Machine Hours	80,000	60,000	10,000	150,000

Calculation of Product Costs Using ABC Allocation Method

Step 1: Identify cost pools

Step 2: Assign overhead costs to cost pools

Step 3: Identify the cost drivers for each activity, and estimate annual activity levels.

Step 4: Compute the predetermined overhead rate for each cost driver.

Activity	Cost Driver	Estimated Overhead Costs	Total Activity	ABC Rate per Activity
Setting Up Machines	Machine setups	\$ 200,000		\$
Purchasing Material	Number of purchase requisitions	500,000		
Inspecting Products	Inspection hours	300,000		
Assembling Products	Number of parts requiring labor	600,000		
Technological Production	Machine hours	900,000		
Total		<u>\$ 2,500,000</u>		

Calculation of Product Costs Using ABC Allocation Method

Step 5: Allocate overhead costs to products to get the total overhead and overhead per unit (two phases).

- **Phase 1:** Determine the expected total overhead cost for each product for each activity.

Activity	Expected Cost Driver Activities			Total
	Solo	Band	Orchestra	
Machine Setups	\$ 80,000 ◆	\$ 60,000	\$ 60,000	\$ 200,000
Number of Purchase Requisitions	250,000 ◆	200,000	50,000	500,000
Inspection Hours	150,000 ◆	135,000	15,000	300,000
Number of Parts Requiring Labor	300,000 ◆	60,000	240,000	600,000
Machine Hours	480,000 ◆	360,000	60,000	900,000
Total Overhead	<u>\$1,260,000</u>	<u>\$815,000</u>	<u>\$425,000</u>	<u>\$2,500,000</u>

Calculation of Product Costs Using ABC Allocation Method

- **Step 5:** Phase 2: Determine the overhead cost **per unit**.

Activity	Solo	Band	Orchestra	Total
Total Overhead				
Number Units	_____	_____	_____	
Overhead per Unit	=====	=====	=====	

- Finally, determine the **per unit** cost:

	Solo	Band	Orchestra
Direct Materials per Unit			
Direct Labor per Unit			
Overhead per Unit			
Cost per Unit via ABC	\$=====	=====	=====

Figure 6.9

Musicality's gross profit **per unit** using activity-based costing:

	Solo	Band	Orchestra
Sales Price			
Cost per Unit (ABC)			
Gross Profit (loss) per Unit	==	==	==

Musicality's gross profit **by product line** using activity-based costing:

	Solo	Band	Orchestra	Total
Number of Units Sold	150,000	110,000	200,000	460,000
ABC Costing				
Sales	\$			
Cost of Goods Sold	_____	_____	_____	_____
Gross Profit (loss)	_____	_____	_____	_____

Figure 6.10

Compare and Contrast Traditional and Activity-Based Costing Systems

Musicality's per unit product cost differences: traditional costing versus activity-based costing

	Solo	Band	Orchestra
Cost per Unit via ABC	\$22.50	\$16.90	\$17.70
Cost per Unit via Traditional	<u>18.50</u>	<u>16.75</u>	<u>20.00</u>
Difference	\$ 4.00	\$ 0.15	\$(2.30)

Compare and Contrast Traditional and Activity-Based Costing Systems

Musicality's gross profit per unit using **activity-based costing**:

	Solo	Band	Orchestra
Sales Price	\$20.00	\$25.00	\$30.00
Cost per Unit (ABC)	<u>22.50</u>	<u>16.90</u>	<u>17.70</u>
Gross Profit (loss) per Unit	<u><u>\$(2.50)</u></u>	<u><u>\$ 8.10</u></u>	<u><u>\$12.30</u></u>

Musicality's gross profit per unit using **traditional allocation**:

	Solo	Band	Orchestra
Sales Price	\$20.00	\$25.00	\$30.00
Cost per Unit (traditional)	<u>18.50</u>	<u>16.75</u>	<u>20.00</u>
Gross Profit per Unit	\$ 1.50	\$ 8.25	\$10.00

The Calculation of Product Costs Using the Activity-Based Costing Allocation Method



Step 1: Identify cost pools

Step 2: Assign overhead costs to cost pools

Step 3: Identify the cost drivers for each activity, and estimate annual activity levels

Step 4: Compute the predetermined overhead rate for each cost driver

Step 5: Allocate overhead costs to products to get the total overhead and overhead per unit

In-class Exercise

Chapter 6, EA 7. Rex Industries has two products. They manufactured 12,539 units of product A and 8,254 units of product B. The data are:

Activity in Cost Pool	Estimated Overhead	Product A	Product B
1	\$45,900	1,000	500
2	67,300	200	800
3	25,200	600	5,400

What is the activity rate for each cost pool?

In-class Exercise

Chapter 6, EA 8. Rex Industries has identified three different activities as cost drivers: machine setups, machine hours, and inspections. The overhead and estimated usage are:

Activity	Overhead per Activity	Annual Usage
Machine setups	\$155,000	4,000
Machine hours	352,800	14,112
Inspections	120,750	3,500

Compute the overhead rate for each activity.

Chapter 6, EA 10. Custom's has three cost pools and an associated cost driver to allocate the costs to the product. The cost pools, cost driver, estimated overhead, and estimated activity for the cost pool are:

Cost Pool	Cost Driver	Estimated Activity per Driver	Estimated Overhead
Material	Material requisitions	14,000	\$ 52,500
Machining	Machine hours	186,400	363,500
Inspection	Number of inspections	10,000	75,000

What is the predetermined overhead rate for each activity?

Chapter 6, EA 9. Custom's makes two types of hats: polyester (poly) and silk. There are two cost pools: setup, with an estimated \$100,000 in overhead, and inspection, with \$25,000 in overhead. Poly is estimated to have 750,000 setups and 170,000 inspections, while silk has 250,000 setups and 80,000 inspections. How much overhead is applied to each product?

Chapter Outline

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6.5 Compare and Contrast Variable and Absorption Costing

Overhead in Traditional versus ABC Costing

	Traditional	ABC
Overhead assigned	Single cost driver	Multiple cost drivers
Optimal usage	When direct labor is a large portion of the product cost	When technology is a large portion of the product cost
Orientation	Cost driven	Process driven

Table 6.4

Compare and Contrast Traditional and Activity-Based Costing Systems

	Traditional Costing	Activity-Based Costing
Advantages	All manufacturing costs are classified as material, labor, or overhead and assigned to products.	There are multiple overhead cost pools, and each has an individual measure of activity.
	All manufacturing costs are considered to be part of the product cost.	The allocation bases, or measures of activity, are often different from those used in traditional allocation.
	There is only one overhead cost pool and a single measure of activity.	Both nonmanufacturing costs and manufacturing costs may be assigned to products.
Disadvantages	The use of the single cost driver does not allocate overhead as accurately as using multiple cost drivers: it may over-allocate overhead to one product and under-allocate overhead to another product.	An ABC system takes much more time and resources to implement and operate.

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Compare and Contrast Variable and Absorption Costing



- **Absorption costing** “absorbs” all of the costs used in manufacturing and includes fixed manufacturing overhead as product costs.
 - Absorption costing is in accordance with GAAP because the product cost includes fixed overhead.
- **Variable costing** considers the variable overhead costs and does not consider fixed overhead as part of a product’s costs.
 - It is not in accordance with GAAP because fixed overhead is treated as a period cost and is not included in the cost of the product.

Example 1 (Figure 6.11: FG Inventory)

Finished goods inventory with absorption costing and variable costing

- Assume a company has fixed overhead of \$12,000 and manufactures 10,000 units. Direct materials cost is \$3 per unit, direct labor is \$15 per unit, and the variable manufacturing overhead is \$7 per unit.

	Absorption	Variable
Materials (\$3 per unit)	\$	
Labor (\$15 per unit)		
Variable Overhead (\$7 per unit)		
Fixed Overhead (\$1.20 per unit)		
Total Finished Goods Inventory	\$ _____	_____

Example 1 (Figure 6.12: Net Income when EI = 0)

Income statement with absorption costing and variable costing:

- Assume each unit is sold for \$33 each.
- all inventory sold, no finished goods inventory

	Absorption	Variable
Sales	\$	
Cost of Goods Sold	_____	_____
Gross Profit	\$	
– Fixed Overhead	_____	_____
Net Income	\$ =====	 =====

Example 1 (Figure 6.13: COGS and Ending Inventory)

Cost of goods sold, ending inventory, and net income with absorption costing and variable costing

- 8,000 units sold and 2,000 units in ending inventory.

	Absorption	Variable
Cost of Goods Sold		
Materials		
Labor		
Variable Overhead		
Fixed Overhead		
Cost of Goods Sold	_____	_____
Ending Inventory		
Materials		
Labor		
Variable Overhead		
Fixed Overhead		
Ending Inventory	_____	_____

	Absorption	Variable
Sales		
Cost of Goods Sold		
Gross Profit		
– Fixed Overhead		
Net Income	=====	=====

Your Turn: Comparing Variable and Absorption Methods

A company expects to manufacture 7,000 units. Its direct material costs are \$10 per unit, direct labor is \$9 per unit, and variable overhead is \$3 per unit. The fixed overhead is estimated at \$49,000. How much would each unit cost under both the variable method and the absorption method?

Chapter 6, EA 14. Cool Pool has these costs associated with production of 20,000 units of accessory products: direct materials, \$70; direct labor, \$110; variable manufacturing overhead, \$45; total fixed manufacturing overhead, \$800,000. What is the cost per unit under both the variable and absorption methods?

In-class Exercise

Chapter 6, EA 15. Using this information from Planters, Inc., what is the cost per unit under both variable and absorption costing?

Production	10,000
Direct materials	\$ 240
Direct labor	280
Variable manufacturing overhead	100
Fixed manufacturing overhead	1,200,000

REVIEW

- With traditional costing, a _____ cost driver, typically direct labor hours, direct labor dollars, or machine hours, is used to apply the predetermined overhead rate.
- Traditional allocation is optimal when the manufacturing process is _____ driven and overhead increases based on traditional activity bases, such as direct labor hours, direct labor dollars, or machine hours.
- With activity-based costing, _____ cost drivers and a detailed break down of estimated overhead into cost pools are used to apply overhead to products based on the _____ of those various activities.
- ABC costing is optimal when the manufacturing process is _____ driven and overhead increases based on various activities that differ for each product.
- Absorption costing assigns _____ manufacturing costs to products, whereas variable costing _____ assigns variable costs to the products.

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